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Palaeomagnetism of limestones from the Gargano Peninsula (Italy), and the implication of these data

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Summary. The platform limestones of Apulia are usually too weakly magnetic for precise measurement. The East Gargano basin is an autochthonous extension of Apulia and incorporates deeper water limestones which, though weakly magnetic (J NRM $\simeq$ 50 nG), can be reliably measured using a cryogenic magnetometer. The magnetization is attributed to the presence of detrital magnetite and the pelagic limestones yield a mean magnetic direction for the Late Cretaceous (Dec. = 335°, Inc. = 38°, $\alpha_{95} = 6.5°$). The circle of confidence associated with this direction overlaps with those associated with Late Cretaceous magnetic directions from Iblei (Sicily) and from the Vicentinian (Southern) Alps. Palaeomagnetic pole positions for Iblei, Gargano/Apulia and the Southern Alps indicate that these three autochthons need not have rotated significantly relative to each other since the Late Cretaceous. An inferred Late Cretaceous pole position for Africa coincides with the pole positions obtained from these Italian data.

Introduction

The Southern Apennines represent part of the deformed Periadriatic continental margin (D'Argenio 1974) which can be subdivided into a number of structural–stratigraphic units (D'Argenio & Scandone 1970; D'Argenio, Pescatore & Scandone 1975; Ippolito *et al.* 1975; Catalano *et al.* 1977). These units have been thrust eastward on to the Apulian platform (Fig. 1) which is generally considered to have acted as the stable autochthonous foreland to Southern Apennine orogeny. Palaeomagnetic results from this area are probably applicable to autochthonous Italy and may be correlated with data from autochthonous sites in Sicily (Iblei) and in the Southern (Vicentinian) Alps. Unfortunately, most of Apulia consists of shallow water platform limestones which are generally too weakly magnetic for precise measurement. However, the Apulian limestones contain an Upper Cretaceous bauxite horizon. Some bauxite samples yield stable, consistent magnetic directions (Channell & Tarling 1975) but these directions are not entirely satisfactory as the magnetization is generally the result of *in situ* growth of pisolites and this pisolitic growth occurs during diagenesis. The magnetization cannot be directly associated with the stratigraphic age of the bauxite, however, flattening of pisolites in a plane parallel to bedding tends to suggest that the pisolites were produced early in diagenesis before the compaction of the sediment was

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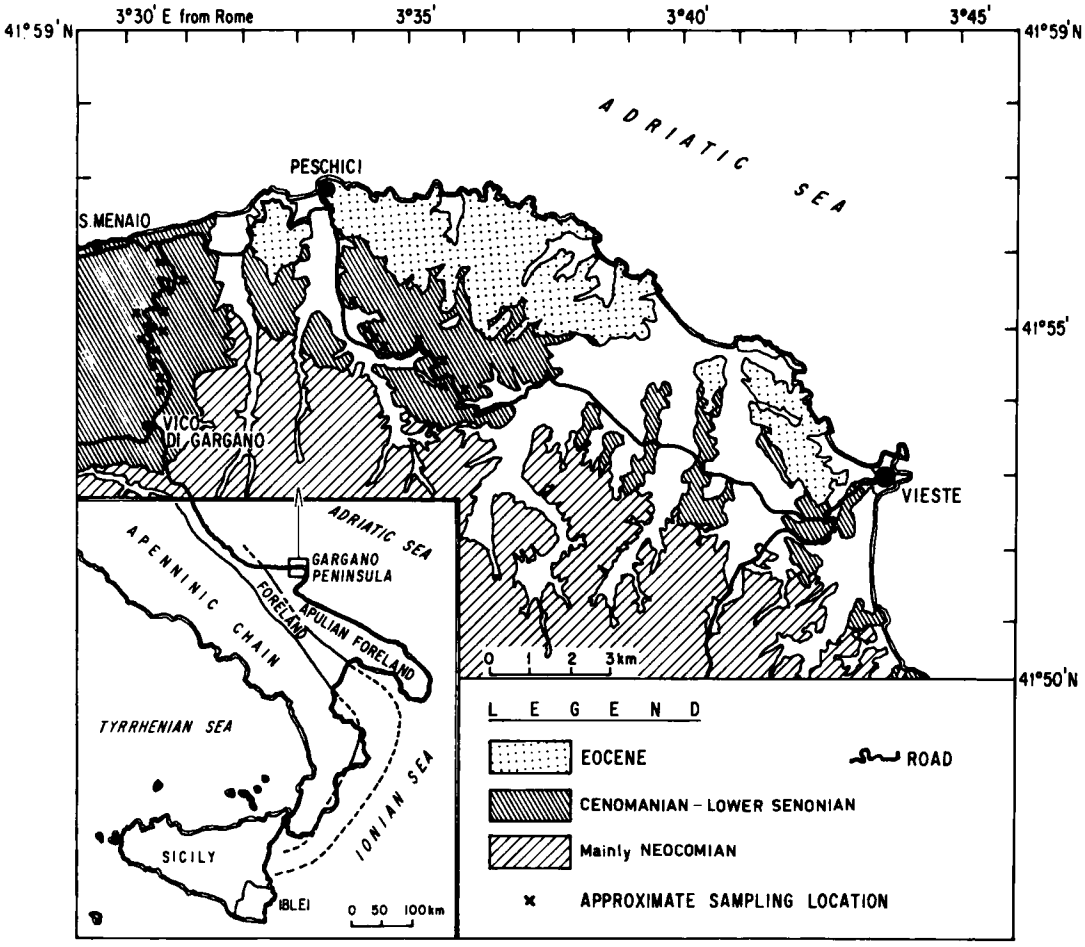


Figure 1. Sketch map of North Gargano sampling locality.

complete. Reliable palaeomagnetic data form the Apulian autochthon are critical for the palaeogeographic reconstruction of the Adriatic region and it was therefore necessary to look for a more amenable rock type for palaeomagnetic study.

The East Gargano Basin (Fig. 1) developed within the Apulian platform carbonates during the Late Jurassic and Early Cretaceous. Fine-grained pelagic limestones were deposited in this basin throughout the Cretaceous and Early Tertiary. These white-to-light-grey pelagic limestones have significantly higher magnetization intensities than the adjacent Apulian platform carbonates. The slower rate of carbonate sedimentation for the pelagic limestones is probably the major factor which accounts for the increased concentration of detrital magnetite, the main contributor to the remanence. The average magnetic intensity, however, of these pelagic limestones (50 nG) is at least an order of magnitude lower than the intensities associated with pelagic limestones (Scaglia) of the same age in Umbria or the Southern Alps. The intensity of the remanent magnetization is such that the measurement of these rocks could only be undertaken using the superconducting ScT magnetometer (Goree & Fuller 1976) which has a noise level of the order of 5 nG.

Two localities in the East Gargano Basin were sampled. Several sites were collected from near Monte S. Angelo on the southern coast of the Gargano Peninsula. At this locality the

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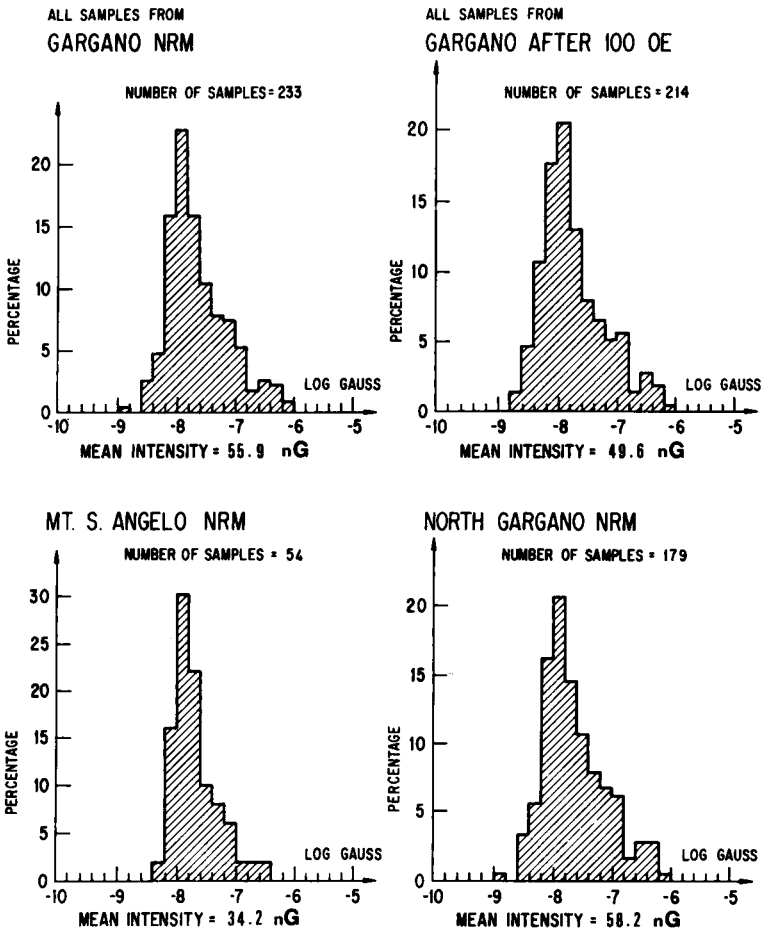


Figure 2. Histograms of magnetization intensity.

limestones are more chalky and their directions usually cannot be measured with sufficient precision. The bulk of the samples (sites 16 to 35, Table 1) were collected from the northern coast of Gargano where the limestones are more silicified and have higher magnetic intensities. Twenty sites were collected along the roads between Vieste and Peschici, and between Vico di Gargano and S. Menaio (Fig. 1). These sites produced 127 drilled cores and one sample was sliced from each core.

Unfortunately, these limestones are almost devoid of the foraminifera (Globotruncana) which have been used so successfully to date the Upper Cretaceous Scaglia limestones in Umbria and the Southern Alps. Radiolarites are common in these sediments but radiolarian stratigraphy is not yet widely developed. However, sites 26–29 yielded poorly preserved Globotruncana which suggest that the sediments belong to the lower part of the Upper Cretaceous (possibly Turonian/Coniacian).

2 Magnetic properties

The limestones are very weakly magnetic. The average NRM intensity from all the sites, including those from Monte S. Angelo, is 55.9 nG (Fig. 2). The average NRM intensity of the sites from Northern Gargano (58 nG) is higher than that of the sites from Monte S. Angelo

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Table 1.

Site	No. of samples	Av. NRM intensity (nG)	NRM Dec. (°)	Inc. (°)	α_{95} (°)	After 100 Oe Dec. (°)	Inc. (°)	α_{95} (°)	After 150 Oe Dec. (°)	Inc. (°)	α_{95} (°)
16	6	30	183.5	−3.1	78.6	*329.5*	*54.8*	*24.3*			
17	7	20	347.5	28.9	14.3	*329.3*	*47.0*	*30.3*	286.4	9.8	61.6
18	4	10	17.5	41.9	52.4	*338.0*	*49.8*	*24.5*			
19	7	20	179.6	−9.1	32.5	*346.0*	*25.0*	*19.0*			
20	4	10	18.6	53.8	38.1	*336.7*	*24.4*	*32.3*			
21	4	20	359.5	40.4	11.3	*6.4*	*39.3*	*27.5*	14.1	24.6	50.9
22	6	10	69.4	20.9	30.1						
23	7	10	88.0	−3.2	35.5						
24	7	10	206.5	6.5	53.6	*351.7*	*37.6*	*33.3*			
25	6	20	338.2	14.7	15.9	340.4	39.7	22.2	*348.7*	*33.1*	*16.3*
26	7	30	347.9	33.5	16.8	324.9	74.7	28.0	*329.0*	*37.2*	*20.8*
27	6	60	338.3	39.3	6.4	*328.4*	*35.8*	*9.3*	325.1	31.8	11.3
28	7	50	338.7	51.4	7.8	*344.7*	*45.5*	*8.8*	332.6	46.9	15.9
29	9	200	346.7	39.3	7.9	*337.8*	*38.6*	*8.6*	341.2	35.9	10.7
30	7	20	349.4	27.0	18.7	*340.4*	*42.0*	*18.8*	325.4	31.1	34.2
31	7	40	333.5	41.1	8.7	*333.2*	*36.9*	*7.7*	322.4	39.4	13.9
32	8	20	321.7	43.2	28.8	*312.7*	*26.1*	*31.8*			
33	7	10	219.2	−16.4	35.8						
34	9	10	326.5	29.5	11.0	341.0	34.3	30.0	*314.9*	*27.0*	*7.9*
35	9	100	319.0	34.7	11.2	308.5	27.9	15.5	*314.3*	*28.9*	*5.8*

Unweighted mean of the 17 (italicised) site mean directions: Dec. = 335.0°, Inc. = 37.8°, K = 31.3, α_{95} = 6.5°.
Mean of the 12 site mean directions for which $\alpha_{95} < 25°$: Dec. = 333.5°, Inc. = 38.4°, K = 40.1, α_{95} = 6.9°.
Mean of the 9 site mean directions for which $\alpha_{95} < 20°$: Dec. = 333.9°, Inc. = 35.4°, K = 40.6, α_{95} = 8.2°.

(34 nG). The Monte S. Angelo sites frequently have intensities uncomfortably close to the noise level of the ScT magnetometer and these sites are not discussed in this paper.

In the case of the sites from Northern Gargano, during alternating field demagnetization the intensity of magnetization decreases and approaches the noise level of the magnetometer. It is necessary to know if a measurement is reliable. The ScT cryogenic magnetometer measures the magnetization along three orthogonal directions at the same time and almost instantaneously, regardless of the intensity of magnetization. The samples were measured in the upright and inverted position, and the empty sample holder was measured before each sample measurement. Following Heller (1977), the six magnetization component readings were combined to produce eight estimates of the remanent magnetization vector. The circular standard deviation (ψ) of the eight directions about their mean was used as a control parameter.

$$\psi = 81\,[(N-1)/(N-R)]^{-1/2} \text{ (degrees)},$$

where N = number of estimates (8) of magnetic moment vector, R = vector sum of the independent estimates of magnetization vector.

For eight estimates of the magnetic moment vector, the value of R which will be exceeded with a probability of 0.05 in the case of a random population ($K = 0$) is 4.48 (Watson 1956). This value of R corresponds to a value of 57.4° for ψ. Hence if $\psi > 57.4$, there is a probability of 0.95 that the eight estimates of the magnetic moment vector are randomly distributed. This criterion is used as a means of rejecting unreliable measurements. In the case of these weakly magnetized limestones, the value of ψ depends on the intensity of the magnetization of the sample (Fig. 3). The critical value of ψ (57.4) is sometimes exceeded after demagnetization at peak fields of 150 Oe. This same rejection criterion is

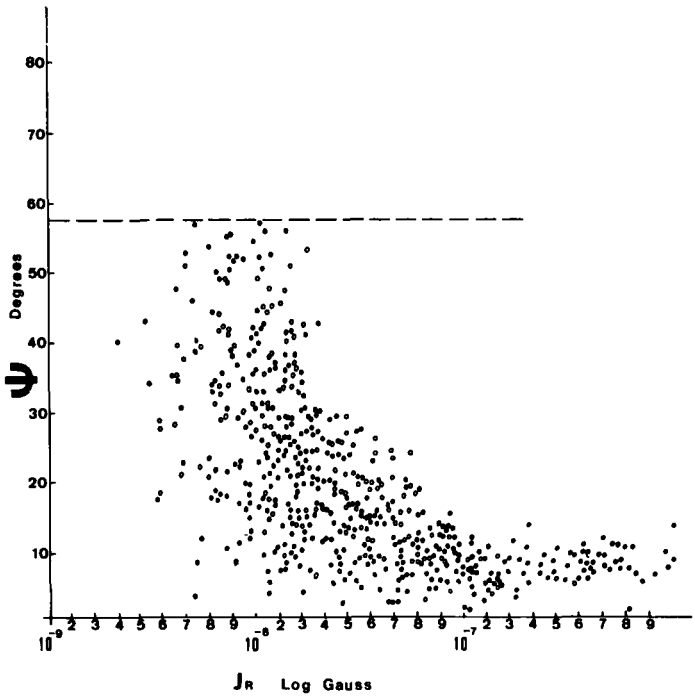


Figure 3. The internal angular standard deviation (ψ) of the magnetic directions about their mean plotted against magnetization intensity.

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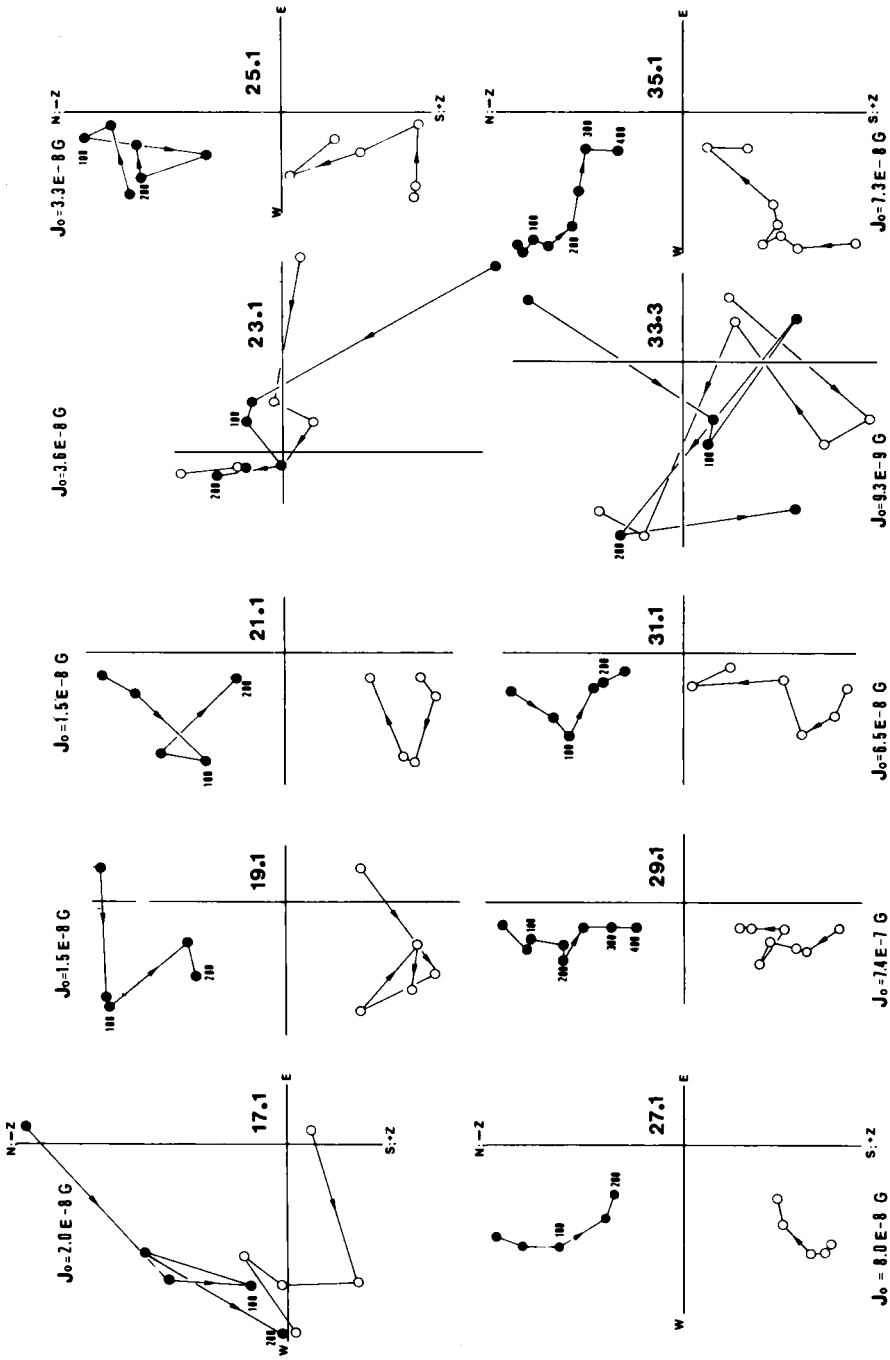


Figure 4. Orthogonal projections (Zijderveld 1967) of the path of the magnetization vector during alternating field demagnetization. Solid symbols represent the projection on to the horizontal plane and open symbols represent the projection on to the vertical (E–W) plane. J_0 is the NRM intensity and the peak alternating fields (Oe) are indicated.

applied to each site. When the vector sum of the sample magnetization vectors (R) within a site is within the range associated with randomness (at the 0.95 probability level) then the site is rejected. Sites 22, 23 and 33 (Table 1) were rejected after demagnetization at peak fields of 100 Oe.

One sample from every other site was demagnetized in steps of 50 Oe up to peak fields of 250–400 Oe. The number of demagnetization steps was limited by the intensity of the individual sample. The af demagnetization curves (Fig. 4) represent reasonable magnetic stability except in the case of 23.1 and 33.1. These two samples have the lowest NRM intensities and the poor stability reflects the lack of precision in measurement rather than differing magnetic properties. The median destructive field varies considerably from sample to sample and often lies between 100 and 200 Oe.

The rate of acquisition of IRM (Isothermal Remanent Magnetization) can give a measure of the coercivity spectrum of the magnetic mineralogy in a rock sample (Dunlop 1972). The three samples which were tested show that saturation magnetization is attained in fields of between 1 and 2 K Oe (Fig. 5). A coercivity spectrum in this range (1–2 K Oe) suggests that magnetite is the dominant magnetic mineral in these white limestones. It is probable that the magnetite occurs as detrital grains which have aligned themselves with the ambient magnetic field either during or shortly after deposition. If so, this magnetization can be associated with the biostratigraphic age of the limestones.

After stepwise demagnetization of pilot samples, the rest of the samples were demagnetized in peak fields of 100 and 150 Oe (Table 1). In some cases the intensity after 150 Oe was such that the values of the measurement internal precision parameter (ψ) exceeded the acceptable maximum. The optimum mean direction for each site was selected by choosing the direction (after demagnetization at either 100 or 150 Oe) which is accompanied by the lower value of α_{95} (cone of 95 per cent confidence).

The site mean directions from the 17 non-random populations of sample magnetization directions (Table 1, Fig. 6) are all significant and should therefore all be considered in the calculation of the formational mean direction. In the case of these very weakly magnetized

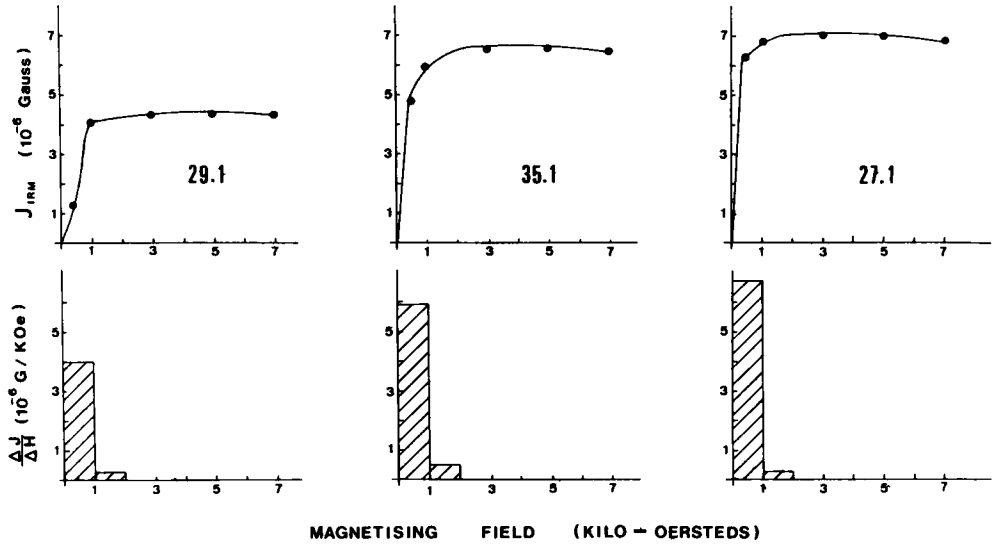


Figure 5. The acquisition curves of isothermal remanent magnetization for three samples from North Gargano.

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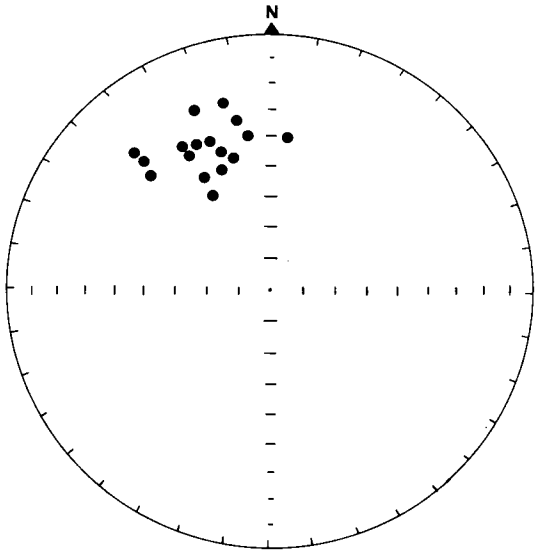


Figure 6. Equal area stereographic projection of the 17 site mean directions from North Gargano.

sediments, the within site precision is very variable and it is unsatisfactory to give equal weight to each site mean direction. It is desirable to weight the site mean directions in favour of those that are better defined. In the palaeomagnetic literature, this 'weighting' is often performed by selecting an arbitrary value of K (precision parameter) or α_{95} (circle of 95 per cent confidence) and rejecting sites which have higher directional scatter. Although this procedure provides a well-documentable selection based on reproducibility, it is somewhat unsatisfactory not only because the choice of K (min) or α_{95} (max) may influence the formational mean direction, but also because information is lost from the data set. In this paper, all the sites which give non-random mean directions at the 0.95 probability level are included in the calculation of the overall mean direction (Dec. = 335.0°, Inc. = 37.8°). In the case of this particular distribution, the elimination of site mean directions with values of α_{95} greater than 25° and 20° does not greatly alter the overall mean direction (Table 1). The application of a weighting factor to each site mean direction enhances the contribution of the better defined site mean directions and gives a slightly different overall mean (Fig. 7). The weighting procedure involved assigning a vector length of (Rw/Nw) to each site mean direction, where Rw is the vector sum of the sample vectors and Nw is the number of samples in the site. Weighting factors are not generally used to produce formational means due to the difficulty of applying Fisherian statistics to weighted mean directions.

3 Implications

This new Upper Cretaceous result from Gargano can be compared with the bauxite result from neighbouring Apulia and with results from other areas in Italy which have been considered autochthonous, namely the Iblean zone of Sicily, Umbria and the Vicentinian Alps (Fig. 7). The age of the Scaglia samples which have been used for palaeomagnetism in Umbria and the Vicentinian Alps is precisely known from the foraminifera (Globotruncana). The Iblean (Sicilian) palaeomagnetic data (Schult 1973; Barberi *et al.* 1974; Gregor, Nairn & Negendank 1975) are based on results from basaltic rocks which give reversed polarity. These basalts are overlain by Senonian rudistid limestones, and Upper Cretaceous Scaglia

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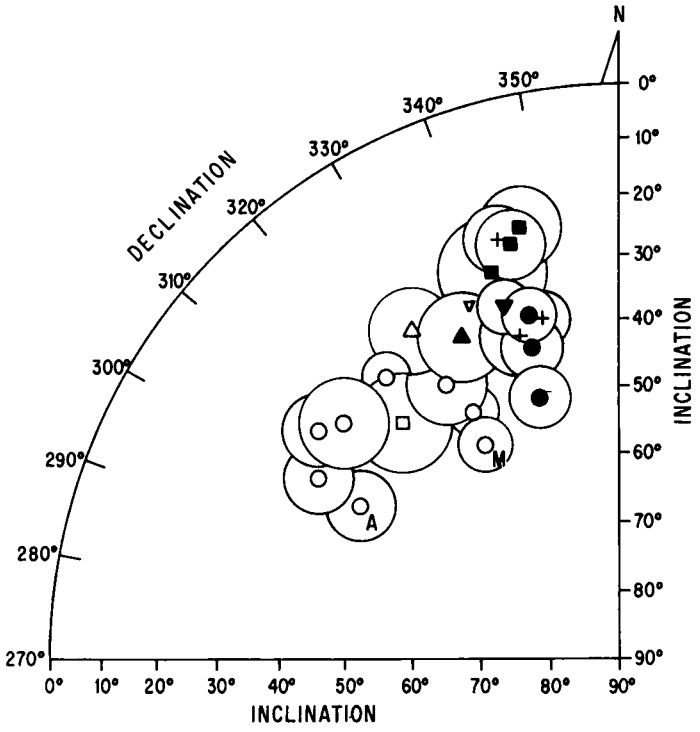


Figure 7.

Locality	Symbol	No. of sites	Age	Reference
Umbria	□	34	Turonian–Eocene	Lowrie & Alvarez 1975
	△	27	Turonian–Eocene	Channell & Tarling 1975
Cagli (Umbria)	○	23	Dated and grouped from Aptian (A) to Maestrictian (M)	Van den Berg *et al.* 1977
Gargano				
Unweighted	▲	17	Turonian/Coniacian (?)	This paper
Weighted	▽			
Apulia	▼	4	Senonian	Channell & Tarling 1975
Iblei	■	19	Senonian	Schult 1973
	■	12	Senonian	Gregor *et al.* 1975
	■	18	Senonian	Barberi *et al.* 1974
Vicentinian Alps	●	2	Late Santonian–Early Campanian	Van den Berg & Wonders 1976
	●	2	Early Santonian	
	●	2	Late Turonian–Coniacian	
	+	(46)	Santonian–Maestrictian	Unpublished data
	+	(25)	Campanian–Maestrictian	Unpublished data
	+	12	Turonian/Eocene	Channell & Tarling 1975

type pelagic calcilutites are known at their base from borehole data. Their negative polarity therefore confines them to the top end (Lower Campanian) of the Cretaceous 'long normal' polarity event. The Apulian bauxites are capped by Senonian limestones (D'Argenio 1970) but the magnetization was produced during diagenesis and its age is not precisely known. As mentioned above, there is also uncertainty concerning the age of the Gargano limestones.

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The circles of 95 per cent confidence associated with the mean directions from Apulia, Gargano and Iblei usually overlap with data from the Vicentinian (Southern) Alps (Fig. 7). The Upper Cretaceous data from the region of the Vicentinian Alps comprise six well-dated sites collected from Turonian to Santonian Scaglia (Van den Berg & Wonders 1976), two Santonian–Maastrictian sections sampled at Castello Tesino (68 samples from 46 beds) and Colli Euganei (62 samples from 26 beds) (Channell, unpublished data), and Scaglia rossa sampled in the region of Schio (Channell & Tarling 1975).

Palaeomagnetic data from the Upper Cretaceous Scaglia of the Umbrian sequence (Lowrie & Alvarez 1975; Channell & Tarling 1975; Van den Berg, Klootwijk & Wonders 1977) deviate from the Southern Alpine data (Fig. 7). This has been interpreted as being partly due to Tertiary fault movement in the Po basin (Van den Berg & Wonders 1976). The Umbrian data of Van den Berg *et al.* (1977) come from near Cagli on the nothern limb of the Umbrian orocline. They recognized an 'African pattern' in the Umbrian data and applied these results to autochthonous Italy. It was proposed that autochthonous Italy, south of the Po basin, moved with Africa until post Early Eocene when a 25°–30° anticlockwise rota-

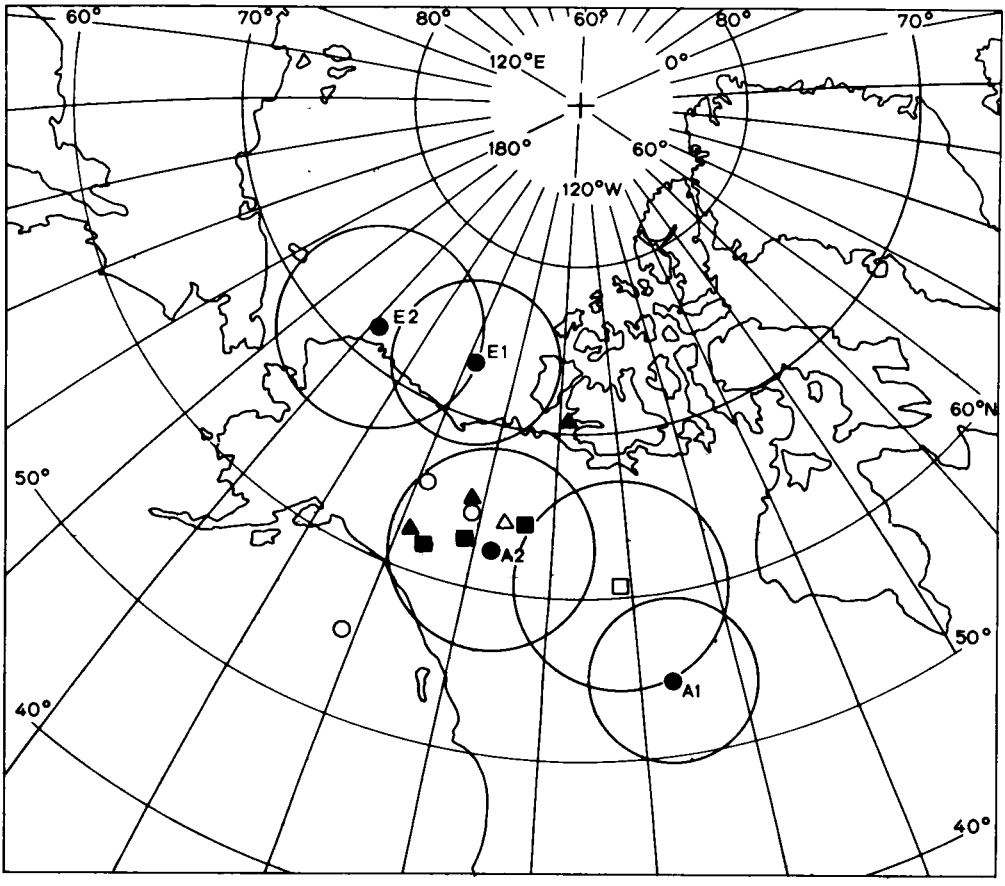


Figure 8. ● Inferred Early Cretaceous (A1) and Late Cretaceous (A2) pole positions for Africa, and inferred Early Cretaceous (E1) and Late Cretaceous (E2) pole positions for Europe (Van der Voo & French 1974). ○ Scaglia limestones, Vicentinian Alps (Channell & Tarling 1975 and unpublished data). ▲ Scaglia limestones, Vicentinian Alps (Van den Berg & Wonders 1976). ■ Basalts, Iblei (Sicily) (Schult 1973, Barberi *et al.* 1974; Gregor *et al.* 1975). △ Bauxites, Apulia (Channell & Tarling 1975). □ Limestones, Gargano (this paper).

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tion of the Italian peninsula occurred relative to Africa. The Southern Alps were considered to have been detached from the rest of autochthonous Italy during the Late Cretaceous and to have undergone a *clockwise* rotation relative to Africa and a later Tertiary *counterclockwise* rotation of the same magnitude such that there is now consistency between Southern Alpine and African data but an offset between Southern Alpine and Umbrian data (Van den Berg & Wonders 1976). Geographically distributed palaeomagnetic sites from Umbria (Lowrie & Alvarez 1975; Channell & Tarling 1975) have been palaeontologically dated, and within individual foraminiferal zones there is indication that the magnetic declination is a function of the geographical position of the site (Channell 1977). If this is the case, the data from Cagli (Van den Berg *et al.* 1977) are not applicable to autochthonous Italy and the more westerly declinations (Fig. 7) are a result of the geographical distribution of the sites and of local tectonics.

The pole positions associated with the mean directions from Iblei, Apulia, Gargano and the Vicentinian Alps (Fig. 8) suggest that these areas have undergone little, if any, relative rotation since the Late Cretaceous. Mesozoic data from Europe and Africa are very scarce and it is therefore very difficult to make a meaningful comparison. The pole positions of Van der Voo & French (1974) for the Atlantic bordering continents have been inferred by using the Atlantic magnetic anomaly data together with available palaeomagnetic data (mainly from North America for the Mesozoic). These inferred pole positions provide the only means of comparison for the Italian pole positions. The Italian data can all be bracketed as Late Cretaceous and the data lie within or close to the circle of 95 per cent confidence associated with the African Late Cretaceous pole position (A2, Fig. 8) inferred by Van der Voo & Freneh (1974).

4 Conclusions

The new result from Gargano is important in the context of the palaeogeography of the Adriatic region. The Iblean zone of Southeast Sicily is generally considered on the basis of facies considerations, palaeomagnetism, and the crustal structure in the Straits of Sicily, to be an autochthonous extension of the African platform. The Venetian Alps, Gargano/Apulia and Iblei may be considered as emergent fragments of the largely submerged autochthonous foreland to Periadriatic orogeny. The consistent Late Cretaceous magnetic directions from these three units suggest little, if any, relative rotation since the Late Cretaceous. These directions give palaeomagnetic pole positions which lie close to an inferred Late Cretaceous pole position for Africa. This tends to suggest, in the light of additional lines of evidence (Channell & Horváth 1976), that the core to Periadriatic orogeny has behaved as a promontory of the African plate.

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